

1. Use trigonometric substitution to find $\int \frac{1}{(\sqrt{x^2+4})^3} dx$.

(Note: You will not receive credit if you use other methods.)

$$\int \frac{1}{(\sqrt{x^2+4})^3} dx = \int \frac{1}{(2\sec\theta)^3} 2\sec^2\theta d\theta$$

$$= \frac{1}{4} \int \frac{1}{\sec\theta} d\theta$$

$$= \frac{1}{4} \int \cos\theta d\theta = \frac{1}{4} \sin\theta + C$$

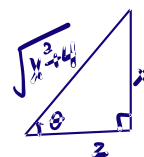
$$= \frac{1}{4} \frac{x}{\sqrt{x^2+4}} + C$$

$$x = 2\tan\theta$$

$$dx = 2\sec^2\theta d\theta$$

$$\begin{aligned} \sqrt{x^2+4} &= \sqrt{4\tan^2\theta+4} = \sqrt{4(\tan^2\theta+1)} \\ &= \sqrt{4\sec^2\theta} = 2\sec\theta \end{aligned}$$

$$\frac{x}{2} = \tan\theta$$



$$\sin\theta = \frac{x}{\sqrt{x^2+4}}$$

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\int \cot x \csc x dx = -\csc x + C, \quad \int \csc^2 x dx = -\cot x + C$$

$$\int \tan x dx = \ln |\sec x| + C, \quad \int \cot x dx = \ln |\sin x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C, \quad \int \csc x dx = \ln |\csc x - \cot x| + C$$